EE3032 Introduction to VLSI—Midterm

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Complete editable reconstruction of seven questions from six green photographed panels, corresponding to three printed pages. The panel crops contain no printed date; 2012 is the archive-folder attribution. Point allocations total 100. This reconstruction contains the question text and figures, without student answer/identity pages.

1. (30 points) Complete the following problems briefly.

(a) During the process of VLSI manufacture, chemical-mechanical polishing (CMP) is performed while a layer of material is grown. What is the purpose of CMP? (3 points)

(b) The layer density rule should be met when you design a nano-scale IC. Why? (3 points)

(c) What is the antenna effect? (3 points)

(d) Draw a CMOS inverter and label the drain and source of the PMOS and NMOS transistors. (2 points)

(e) Define the low and high noise margins of a CMOS inverter using the voltage transfer curve of the inverter. (3 points)

(f) List two possible approaches for alleviating the effect of crosstalk on two neighboring parallel interconnects. (4 points)

(g) List three major components of the load capacitance of a CMOS logic gate in a CMOS circuit. (3 points)

(h) How do you size the N-type and P-type MOS transistors of an inverter such that the output voltage of the inverter is $V_{DD}/2$ when the input voltage is $V_{DD}/2$? (3 points)

(i) List two types of FinFET structures. (3 points)

(j) The delay of a long interconnection can be estimated by $k\ell^2$, where k is a constant. What approach can be used to reduce the delay of interconnection? Explain your answer. (3 points)

2. (15 points)

Figure 1 shows a transistor-level schematic realizing a Boolean function f.

(a) Assume that the sizes of all PMOS transistors and NMOS transistors are the same and the turn-on resistance of PMOS and NMOS transistors is R_p and R_n, respectively. Also, parasitic capacitances of nodes 1, 2, 3, and 4 are C_1, C_2, C_3, and C_4, respectively, and $C_1 = C_2 = C_3 = C_4 = C$. If functional inputs (XYZ) of the circuit are $\{010, 100, 110, 000, 101, 111\}$, calculate the worst-case rise time and fall time of the circuit using the Elmore RC delay model. (10 points)

(b) Assume that the voltage swings at nodes 1, 2, 3, and 4 are V_1, V_2, V_3, and V_4. Also, the switching activities of nodes 1, 2, 3, and 4 are $\alpha_1, \alpha_2, \alpha_3$, and α_4. If the logic operates at frequency f, express the internal cell power of the logic gate. (5 points)

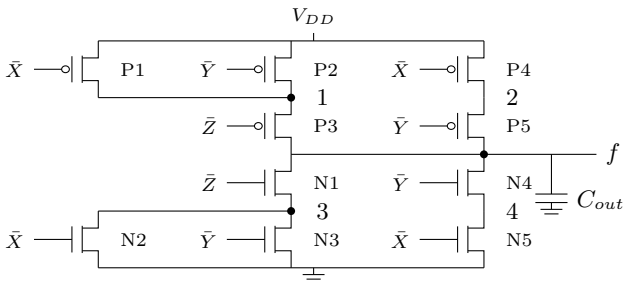


Figure 1: Transistor-level schematic.

3. (10 points)

Figure 2(a) depicts a CMOS logic circuit in which the sizes of NMOS and PMOS for implementing each gate are the same. Assume that a dual-V_t process can be used to implement the CMOS circuit. Draw the transistor schematic diagram of the CMOS circuit in terms of the symbols shown in Fig. 2(b) such that the leakage power of the CMOS circuit is minimal without sacrificing the speed of the CMOS circuit.

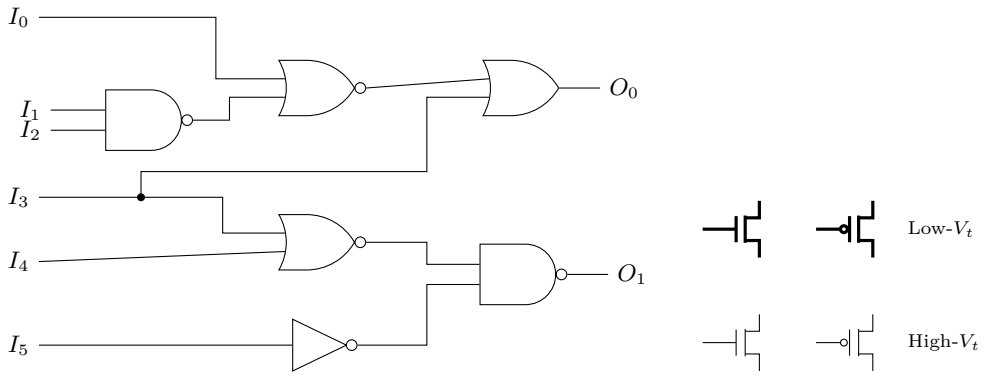


Figure 2: (a) A CMOS circuit (b) High- and low-V_t transistor symbols.

4. (10 points)

Figure 3 shows a pass-transistor logic circuit. Assume that the supply voltage $V_{DD} = 1.8$ V and the threshold voltage of the NMOS transistor is 0.4 V.

(a) Draw the corresponding waveforms at outputs Y and Z if the signal waveform shown in the leftmost figure is applied to input X. Note that you should label the largest and lowest values of voltage at the waveforms. (4 points)

(b) Assume that the body effect of the transistors can be neglected. Also, the signal waveform shown in the leftmost figure is applied to input X.
i. Among the transistors $m1, m2, m3, m4, m5$, and $m6$, identify the transistors which can be operated in saturation and linear regions, and identify the transistor(s) which can be operated in the linear region only. Explain your answers. (6 points)

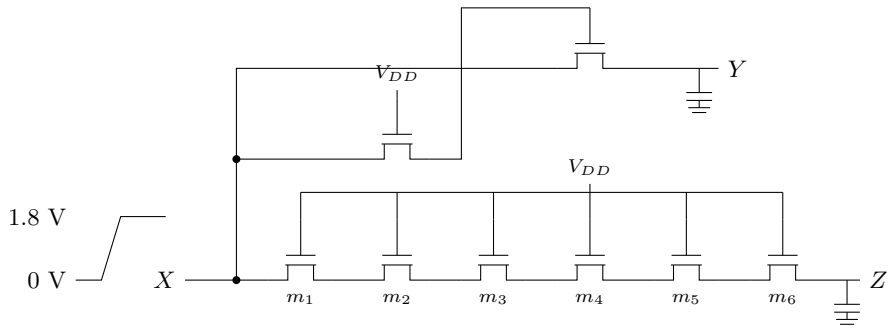


Figure 3: A pass-transistor logic circuit. The upper pass gate receives X as data and a degraded copy of X as its gate; it is not a two-device series data chain.

5. (10 points)

Figure 4 depicts an inverter chain driving a capacitive load C_L. Assume that the three inverters have the same size and every inverter has a minimum-size NMOS and the ratio of PMOS to NMOS is α. Also, the t_r and t_f of an inverter driving a load C_L can be estimated by $t_r = R_p C_L$ and $t_f = R_n C_L$, where R_p and R_n are the turn-on resistances of the PMOS and NMOS of the inverter. Let $\mu_n = 3\mu_p$ and the turn-on resistance and gate capacitance of the minimum-size NMOS be R and C, respectively. Derive the value of α such that $T = 8RC$ and the average switching power contributed by the first two inverters is minimal.

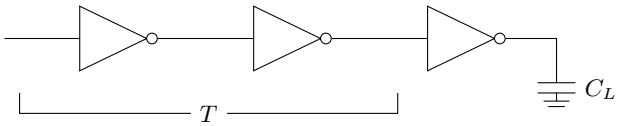


Figure 4: An inverter chain; T brackets the first two inverter delays.

6. (10 points)

Realize the following Boolean functions using static CMOS logic. You should show the transistor diagram.

(a) $f(w, x, y, z) = \overline{w(x + y + z)}$. (5 points)

(b) $f(x, y) = \bar{x}y + x\bar{y}$. (5 points)

Suppose that only true literals are available. Your implementation should use the minimal number of transistors.

7. (15 points)

A full adder is a device that accepts three inputs x, y, and c_{in} and produces two outputs s_{out} and c_{out}, where x, y, c_{in}, s_{out}, and c_{out} denote the addend, augend, carry input, sum, and carry output, respectively.

(a) Derive the truth table for this full adder. (5 points)

(b) From the truth table, derive the s_{out} and c_{out} Boolean functions. (5 points)

(c) Realize it using CMOS logic circuit. (5 points)

Original panels carry the course title and printed page identifiers 3-1, 3-2 and 3-3. No student annotations are added to the reconstructed prompts.